

No. _____
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1. (1) 消左递归 $S \rightarrow a/\wedge/T$
 $T \rightarrow ST'$
 $T' \rightarrow ,ST'/\epsilon$

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P(S)
Begin
  if ch == 'a' then
    match('a')
  else if ch == '^' then
    match('^')
  else if ch == ',' then
    begin
      match(',')
      P(T)
      match(',')
    end
  else
    error()
  end
end
P(T)
begin
  P(S)
  P(T')
end

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P(T')
begin
  if ch == 'a' then
    begin
      match(',')
      P(S)
      P(T')
    end
  end
end

```

(2) $FIRST(S) = \{a, \wedge, (\}$

$FIRST(T) = FIRST(S)$

$FIRST(T') = \{ , , \epsilon \}$

$FOLLOW(S) = \{ , ,) , \$ \}$

$FOLLOW(T) = \{) \}$

$FOLLOW(T') = \{) \}$

对于 $S \rightarrow a/\wedge/T$, $FIRST$ 集合两两互不相交

对于 $T \rightarrow ST'/\epsilon$, $FIRST(ST')$ 与 $FIRST(T')$ 不相交
 是 LL(1) 的

end

a ^ () , \$

S $S \rightarrow a$ $S \rightarrow \wedge$ $S \rightarrow (T)$

T $T \rightarrow ST'$ $T \rightarrow ST'$ $T \rightarrow ST'$

T' $T' \rightarrow \epsilon$ $T' \rightarrow ,ST'$

$$2. (1) \text{FIRST}(P) = \{*, a, b, \wedge\}$$

$$\text{FOLLOW } E$$

$$\text{FOLLOW}(P) = \{\$, \wedge\}$$

$$\text{FIRST}(F') = \{*, \varepsilon\}$$

$$\text{FOLLOW}(E') = \text{FOLLOW}(E)$$

$$\text{FIRST}(F) = \text{FIRST}(P)$$

$$\text{FOLLOW}(T) = \{+\} \cup \text{FOLLOW}(T') \cup \text{FOLLOW}(E)$$

$$\text{FIRST}(T') = \text{FIRST}(F)$$

$$\text{FOLLOW}(T') = \text{FOLLOW}(T) = \{+, \$, \wedge\}$$

$$\text{FIRST}(T) = \text{FIRST}(T') \cup \{\varepsilon\}$$

$$\text{FOLLOW}(F) = \{\$, a, b, \wedge\} \cup \text{FOLLOW}(T)$$

$$\text{FIRST}(E') = \{+, \varepsilon\}$$

$$\text{FOLLOW}(F') = \text{FOLLOW}(F) = \{\$, a, b, \wedge, +, \$, \wedge\}$$

$$\text{FIRST}(E) = \text{FIRST}(T)$$

$$\text{FOLLOW}(P) = \{*, (, a, b, \wedge, +, \$, \wedge\}$$

$$(2) \textcircled{1} E' \rightarrow +E/\varepsilon \quad \text{FIRST}(+E) \cap \text{FOLLOW}(E') = \emptyset$$

$$\textcircled{2} T' \rightarrow T/\varepsilon \quad \text{FIRST}(T) \cap \text{FOLLOW}(T') = \emptyset$$

$$\textcircled{3} F' \rightarrow *F'/\varepsilon \quad \text{FIRST}(*F') \cap \text{FOLLOW}(F') = \emptyset \quad \text{是 LL(1)}$$

$$\textcircled{4} P \rightarrow (E)/a/b/\wedge \quad \text{FIRST} \text{ 集合两两互不相交.}$$

$$(3) \quad a \quad b \quad \wedge \quad + \quad * \quad (\quad) \quad \$$$

$$E \quad E \rightarrow TE' \quad E \rightarrow TE' \quad E \rightarrow TE' \quad E \rightarrow TE'$$

$$E' \quad E' \rightarrow \varepsilon \quad E' \rightarrow \varepsilon \quad E' \rightarrow \varepsilon \quad E' \rightarrow \varepsilon \quad E' \rightarrow \varepsilon$$

$$T \quad T \rightarrow FT' \quad T \rightarrow FT' \quad T \rightarrow FT' \quad T \rightarrow FT'$$

$$T' \quad T' \rightarrow T \quad T' \rightarrow T \quad T' \rightarrow T \quad T' \rightarrow T \quad T' \rightarrow T \quad T' \rightarrow \varepsilon \quad T' \rightarrow \varepsilon \quad T' \rightarrow \varepsilon$$

$$F \quad F \rightarrow PF' \quad F \rightarrow PF' \quad F \rightarrow PF' \quad F \rightarrow PF'$$

$$F' \quad F' \rightarrow \varepsilon \quad F' \rightarrow \varepsilon \quad F' \rightarrow \varepsilon \quad F' \rightarrow \varepsilon \quad F' \rightarrow \varepsilon \quad F' \rightarrow *F \quad F' \rightarrow \varepsilon \quad F' \rightarrow \varepsilon \quad F' \rightarrow \varepsilon$$

$$P \quad P \rightarrow a \quad P \rightarrow b \quad P \rightarrow \wedge \quad P \rightarrow (E)$$

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(4)

P(E)

begin

P(T)

P(E')

end

P(E')

begin

if ch == '+' then

begin

match('+')

P(E)

end

end

P(T)

begin

P(F)

P(T')

end

P(T')

begin

if ch in { ' ', a, b, \ } then

0 P(T)

else if not (ch in { '+', ' ', '\$' }) then

Campus Error()

end

P(F)

begin

P(P)

P(F')

end

P(F')

begin

if ch == '*' then

begin

match('*')

P(F')

end

end

P(P)

begin

case ch of

'(' : begin match('('); P(E); match(')'); end

'a' : match('a')

'b' : match('b')

'\n' : match('\n')

end

2. (1) $FIRST(S) = \{a, b\}$ $FOLLOW(S) = \{\$ \}$

$FIRST(A) = \{a, \epsilon\}$ $FOLLOW(A) = \{b\}$

$FIRST(B) = \{b, \epsilon\}$ $FOLLOW(B) = \emptyset$

因为 $FIRST(A) \cap FOLLOW(A) = \emptyset$

$FIRST(B) \cap FOLLOW(B) = \emptyset$

所以是 LL(1)

(2). $FIRST(A) = \{a, b, \epsilon\}$ $FOLLOW(A) = \{b\}$

$FIRST(B) = \{b, \epsilon\}$ $FOLLOW(B) = \{b\}$

由于 $FIRST(B) \cap FOLLOW(B) = \{b\} \neq \emptyset$

不是 LL(1)

4. $FIRST(E) = \{-, \epsilon, id\}$ $FOLLOW(E) = \{\$, \epsilon\}$

$FIRST(E_t) = \{-, \epsilon\}$ $FOLLOW(E_t) = \{\$, \epsilon\}$

$FIRST(V) = \{id\}$ $FOLLOW(V) = \{-, \$, \epsilon\}$

$FIRST(V_t) = \{(\epsilon, \epsilon\}$ $FOLLOW(V_t) = \{-, \$, \epsilon\}$

	id	-	(	)	\$
E	$E \rightarrow EV_t$	$E \rightarrow -E$	$E \rightarrow (E)$		
E_t		$E_t \rightarrow -E$		$E_t \rightarrow \epsilon$	$E_t \rightarrow \epsilon$
V	$V \rightarrow idV_t$				
V_t		$V_t \rightarrow \epsilon$	$V_t \rightarrow (E)$	$V_t \rightarrow \epsilon$	$V_t \rightarrow \epsilon$

	id--id((id))\$
1 \$E	id--id((id))\$
2 \$E+V	id--id((id))\$
3 \$E+V+id	id--id((id))\$
4 \$E+V+	--id((id))\$
5 \$E+	--id((id))\$
6 \$E-	--id((id))\$
7 \$E	--id((id))\$
8 \$E-	--id((id))\$
9 \$E	id((id))\$
10 \$E+V	id((id))\$
11 \$E+V+id	id((id))\$
12 \$E+V+	((id))\$
13 \$E+)E(	((id))\$
14 \$E+)E	((id))\$
15 \$E+)E(	((id))\$
16 \$E+)E	id))\$
17 \$E+)E+V	id))\$
18 \$E+)E+V+id	id))\$
19 \$E+)E+V+	)\$
20 \$E+)E+	)\$
21 \$E+))	)\$
22 \$E+)	)\$
23 \$E+	\$
24 \$	\$